

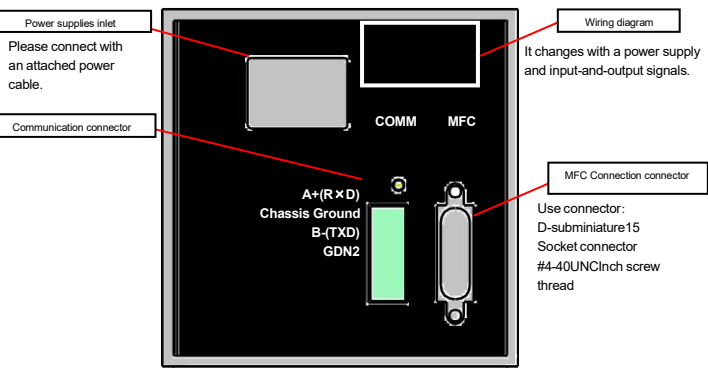
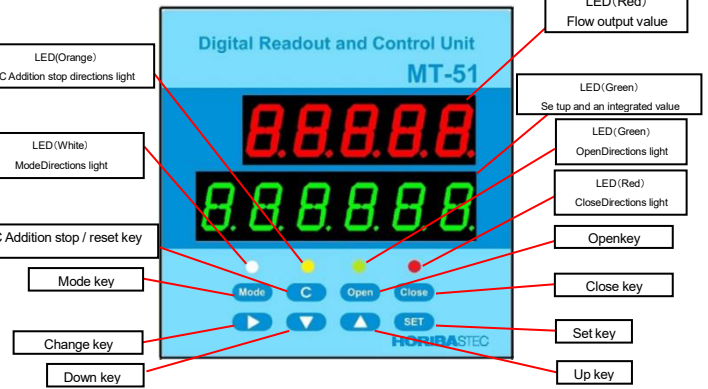
DIGITAL READOUT AND CONTROL
MT-51

CODE: CSMS5101 ver1.3

1.Specifications

Model		MT-51				
Electric supply source (exchange)		AC100~240V 50/60Hz				
Power outlet (direct current)	±15V ±2.5% +15VOutput current max=150mA -15VOutput current max=350mA	+24V DC(22.5~25VDC) Output current max=350mA				
Data backup	Data storage is possible to EEPROM					
Input signal	0~5VDC	0~5VDC	1~5VDC	0~10VDC	4~20mA	
Output signal	0~5VDC	0~5VDC	1~5VDC	0~10VDC	4~20mA	
MFC connection connector	Dsub 15pin female					
AD translation accuracy	16BIT					
Display	LED (Red)	An output value/integrated value Maximum display 4 1/2 beam				
	LED (green)	Set value Maximum display 6 beam				
	Representational accuracy	±0.1%				
Addition maximum	9999 × 10 ⁶					
Addition display	Decimal display:6 beam or Index display:4 beam of cardinal numbers, and double figures index					
Addition accuracy	±1%					
Digital communications	RS-485 or RS232 Modbus-RTU					
Transmission speed	9600(normal) , 19200, 4800bps					
Operation temperature	0~50℃					
Size	96(Length) × 96(Width) × 156(Depth)					
Weight	770g					

2.Operating Panel Side



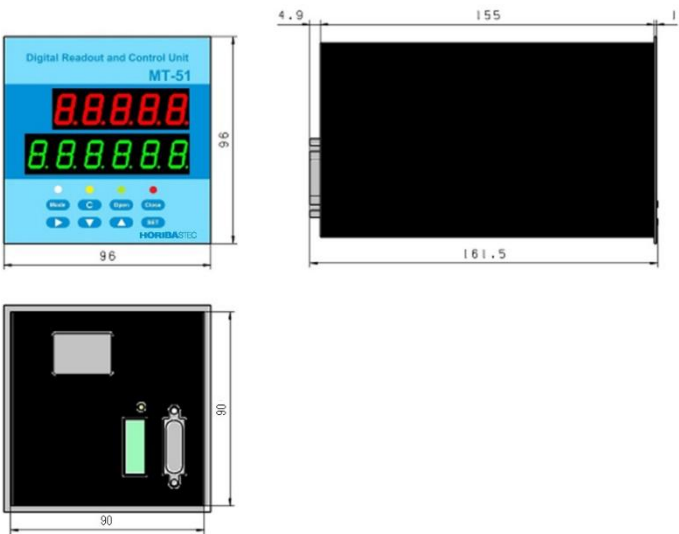
3. Connector Side

Analog use connector: D-subminiature15 socket connector (#4-40UNC inch screw thread)
1)±15VSPEC 2)+24VSPEC

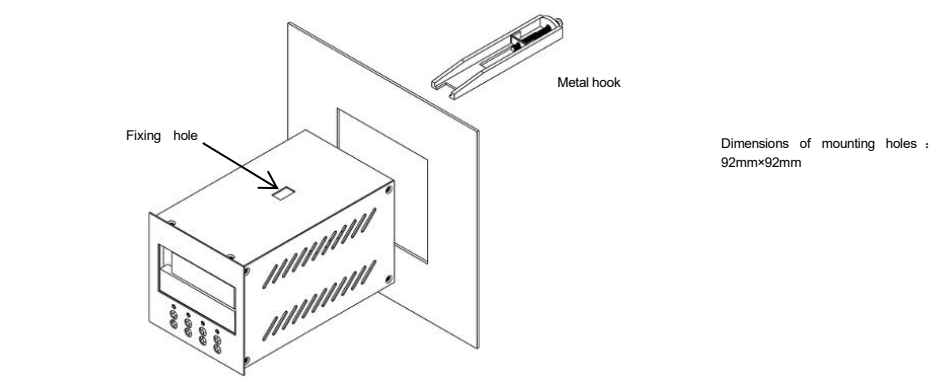
Pin No	Signal
1	N.C.
2	N.C.
3	Signal Common
4	Set input(0-5V)
5	Valve override
6	Signal Common
7	Signal output(0-5V)
8	Ground
9	N.C.
10	N.C.
11	Power supply (-15VDC)
12	Power supply (-15VDC)
13	N.C.
14	Power supply (-15VDC)
15	Power supply (+15VDC)

Pin No	Signal
1	Signal Common
2	Signal output(4-20mA)
3	N.C.
4	Forced valve close signal
5	Power supply(+24VDC)
6	N.C.
7	Forced valve open signal
8	Set input(4-20mA)
9	Power Common
10	Signal Common
11	N.C.
12	N.C.
13	N.C.
14	Ground
15	N.C.

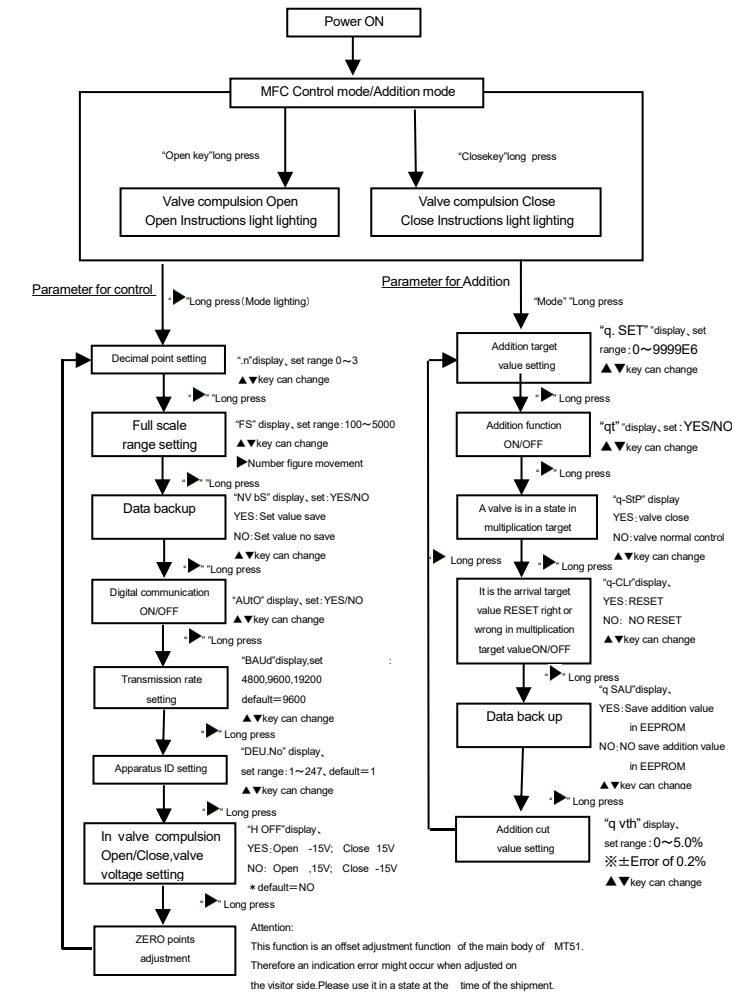
4. External Dimension Drawing



5. Fixing method



6.State Transition Diagram

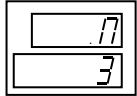


Attention: It goes back up in a control mode from each setting mode when it is not operated at the time of each setting more than ten seconds.

7. Initial setting

7-1 Setting of the decimal point position

Entering full scale range setting mode. Up down key ▼▲ can change value



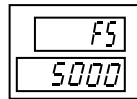
Only a decimal point position flashes on and off. Change range: 0~3

7-2 Full scale value setting

Setting full scale value (equivalent to full scale value of the mass flow controller).

Please set the full scale to a model becoming the use than the following procedures.

Entering the full scale range setting mode, ►key can change beam. Up down key ▼▲ can change value.



A change beam flashes on and off. Full scale range range: 100~5000

8. Explanation of the MFC control mode

MFC control operation mode is a mode controlling with an operation button of the entire surface.

Flow quantity control, valve compulsion are open and can perform valve forced closing.

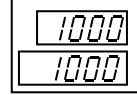
Setting flow quantity is displayed when you input a flow quantity setting signal after power supply injection.

But there are not relations with the true flow quantity, please warn the flow quantity output indication value when a mass flow controller is not connected.

8-1 Flow rate control

In case of flow rate setting, Up down key ▼▲ is pressed ("Mode" Lights out--Lighting) It decides a flow quantity set point. In case of setting value beam is changed, ►key can change The flow quantity signal which was set because setting flow quantity pushes the "SET" key after a decision is input.

►key can change.



Out pu flow rate

Input flow rate

"The SET" is a setting key, and setting is settled by pushing "the SET" after setting. Setting is canceled when I push "Mode" in the middle of setting and returns to a value before the setting.

8-2 Valve forced opening

Open turns on by pushing "Open" of the front valve and becomes the valve forced open state.

"By long pressing "Open", "Open" become Lights out and valve Open cancel.

Priority is valve open signal whatever it input some setting.

8-3 Valve forced closing

Close turns on by pushing "Close" of the front valve and becomes the valve compulsion closing state.

By long pressing "Close", "Close" become Lights out and valve Close cancel.

Priority is valve open signal whatever it input some setting.

9. Explanation of MFC addition mode

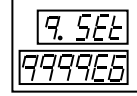
MFC control operation mode is addition mode with an operation button of the entire surface

It performs flow addition, stop, reset.

9-1 Addition target setting

By long pressing "Mode", enter addition parameter mode and up down key ▼▲ can set addition target setting

►key can change beam

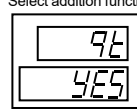


Set range: 0~9999E6

9-2 Addition start

By long pressing "Mode", enter addition parameter mode. ►key long pressing and up down key ▼▲

Select addition function "YES" after that, press "SET" key, addition start



9-3 Addition stop/reset

"C" LED turns on when I push the "C" key during addition, and addition stops. It becomes "C" LED lights out when I push the "C" key again, and addition is started.

When a horizontal piece of timber does "C" key, the addition function is canceled, and the numerical value returns to 0. Because a addition value becomes 0 when the head pushes the "C" key at the time of multiplication tentativeness, "C" LED pushing C "key, please start addition in a state of lights out.

10. Communication function

10-1. Basic information

Communication protocol: Modbus RTU,

Communication level: RS232 or RS485,

Parity : EVEN

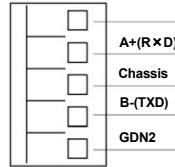
Character bit : 8bit

Stop bit : 1bit

Error detect in : CRC-16

Transmission rate : 4800,9600,19200bps

10-2. Connection



A+(RxD) : RS485 A+, RSRS232 Rxd

Chassis : Ground

B-(TXD) : RS485 B-, RSRS232 Txd

GND2 : COM

10-3. Explanation of each order

10-3-1 Read out flow value

Read out flow value	Item	Function number	Function code	Relativity address	Read out word number	CRC
	Order message	**	03	0010	0001	***
	Length of data	1Byte	1Byte	2Byte	2Byte	2Byte
	Item	Function number	Function code	Read out byte number	Data	CRC
	Respond message	** (1Byte)	03 (1Byte)	02	Dddd(short)	***
	Length of data	1Byte	1Byte	1Byte	2Byte	2Byte

10-3-2 Read out addition value cardinal number

Read out addition value cardinal number	Item	Function number	Function code	Relativity address	Read out word number	CRC
	Order message	**	03	0018	0001	****
	Length of data	1Byte	1Byte	2Byte	2Byte	2Byte
	Item	Function number	Function code	Read out byte number	Data	CRC
	Respond message	** (1Byte)	03 (1Byte)	02	Dddd(ushort)	****
	Length of data	1Byte	1Byte	0Byte	2Byte	2Byte

10-3-3 Read out addition index

Read out addition index	Item	Function number	Function code	Relativity address	Read out word number	CRC
	Order message	**	03	0019	0001	****
	Length of data	1Byte	1Byte	2Byte	2Byte	2Byte
	Item	Function number	Function code	Read out byte number	Data	CRC
	Respond message	** (1Byte)	03 (1Byte)	02	Dddd(us hort)	****
	Length of data	1Byte	1Byte	1Byte	2Byte	2Byte

10-3-4 Read out function number

Read out function number	Item	Function number	Function code	Relativity address	Read out word number	CRC
	Order message	**	03	0033	0001	***
	Length of data	1Byte	1Byte	2Byte	2Byte	2Byte
	Item	Function number	Function code	Read out byte number	Data	CRC
	Respond message	** (1Byte)	03 (1Byte)	02	00dd(usort)	***
	Length of data	1Byte	1Byte	1Byte	2Byte	2Byte

10-3-5 Write in flow value

Write in flow value	Item	Function number	Function code	Relativity address	Write in word number	CRC
	Order message	**	06	0011	Dddd(ushort)	****
	Length of data	1Byte	1Byte	2Byte	2Byte	2Byte
	Item	Function number	Function code	Relativity address	Data	CRC
	Respond message	** (1Byte)	06	0011	dddd	****
	Length of data	1Byte	1Byte	2Byte	2Byte	2Byte

10-3-6 Write in addition value cardinal number

Write in addition value cardinal number	Item	function number	Function code	Relativity address	Write in word number	CRC
	Order message	**	06	001E	Dddd(ushort)	****
	Length of data	1Byte	1Byte	2Byte	2Byte	2Byte
	Item	Function number	Function code	Relativity address	Data	CRC
	Respond message	** (1Byte)	06	001E	dddd	****
	Length of data	1Byte	1Byte	2Byte	2Byte	2Byte

10-3-7 Write in addition index

Write in addition index	Item	Function number	Function code	Relativity address	Write in word number	CRC
	Order message	**	06	001D	Dddd(ushort)	****
	Length of data	1Byte	1byte	2byte	2Byte	2Byte
	Item	Function number	Function code	Relativity address	Dala	CRC
	Respond message	** (1Byte)	06	001D	dddd	****
	Length of data	1Byte	1byte	2byte	2Byte	2Byte

10-3-8 Write in Bit data

Write in Bit data	Item	Function number	Function code	Relativity address	Write in specified state	CRC
	Order message	**	05	0003	dd00	****
	Length of data	1Byte	1byte	2byte	2Byte	2Byte
	Item	Function number	Function code	Relativity address	Data	CRC
	Respond message	** (1byte)	06	0003	dd00	****
	Length of data	1Byte	1byte	2byte	2Byte	2Byte

10-3-9 Detail of bit data

Item	0x8	0x7	0x6	0x5	0x4	0x3	0x2	0x1	0x0
R/W or R	R/W	R	R/W	R/W	–	R/W	R/W	R/W	R/W
Meaning	Addition stop	Alarm	Unit	Addition reset	–	Set switch	Open	Control	Close
Setting	1: Validity 0: Invalidity	1: Validity 0: Invalidity	–	1: Validity 0: Invalidity	–	1: Digitak set 0: Button set	1: Validity Close priority		
Default	0	0	0	0	0	By Panel setting value			

11.Factory Settings

Each Functions of the factory settings are as bellow.

Control mode parameter

- Decimal point setup: 1
- Full-scale range setup: 500
- Data back up: YES
- Digital communication: NO
- Transmission speed setup: 9600
- Apparatus ID setup: 1
- Valve voltage setup at the time of valve compulsion Open/Close.: NO

Addition mode parameter

- Addition desired value setup: 9999E6
- Addition function (ON/OFF): NO
- Valve state to an addition desired value at the time of attainment: NO
- Desired value RESET propriety to an addition desired value at the time of attainment: NO
- Data back up: NO
- Addition cut value setup: 0.0

12.Product warranty

12-1 Period

The product is warranted for one (1) year (parts and labor) from date of shipment. Repair will be provided free of charge during this period if the product is returned to HORIBA STEC or authorized service representative with a description of the problem. HORIBA STEC is not responsible for damage due to customer neglect or improper operation of this product.

12-2 Scope

Warranty coverage is restricted to this product only. HORIBA STEC is not responsible for damage to other components due to improper operation of this product

12-3 Warranty

Replacement parts are warranted for ninety (90) days or the remainder of the warranty period (whichever is longer).

12-4 HORIBA STEC is not responsible for damage ue to:

- a) Natural disasters
b) Miss-operation or abuse of this product
c) Operation or storage in an unsuitable environment
d) Operation outside of the rated specifications
e) Unauthorized alterations or retrofits to this product
- Repair expense with / without charge is to be determined as examination and / or disassembly of the returned

*For questions of service please contact

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URL <http://www.horiba-stec.jp/>

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40, Beiyuanlu, Chouyou-ku Beijing China
URL <http://www.horibaprecision.com>

Appendix : China RoHS

Meaning of Marking

标记的意义

マークの意味


 This marking is applied to electric and electronic products sold in the People's Republic of China. The figure at the center of the marking indicates the environmental protection use period in years. (It does not indicate a product guarantee period.) It guarantees that the product will not cause environment pollution nor serious influence on human body and property within the period of the indicated years which is counted from the date of manufacture as far as the safety and usage precautions for the product are observed. Do not throw away this product without any good reason.

本マークは、中華人民共和国で販売される電気電子製品に適用され、マークの中央の数字は環境保護使用期限の年数を意味します(製品の品質保証期間を示すものではありません)。この製品に関する安全や使用上の注意をお守り頂く限り、製造日から起算するこの年限内では、環境汚染や人体や財産に深刻な影響を及ぼすことはありません。本製品をみだりに廃棄しないでください。

Name and amount of hazardous substance used in a product

产品中有害物质的名称及含量

部件名称 Unit name	有害物质 Hazardous substance					
	铅 Lead (Pb)	汞 Mercury (Hg)	镉 Cadmium (Cd)	六价铬 Hexavalent chromium (Cr (VI))	多溴联苯 Polybromo- biphenyl (PBB)	多溴二苯醚 Polybromo- diphenyl ether (PBDE)
	外壳 Case	×	×	×	×	×
机械零部件 Machine parts	×	×	×	×	×	×
电路板 Printed board	×	×	×	×	×	×

本表格依据 SJ/T 11364 的规定编制。
This form is prepared in accordance with SJ / T 11364.

○: 表示该有害物质在该部件所有均质材料中的含量均在 GB/T 26572 规定的限量要求以下。
Denotes that the amount of the hazardous substance contained in all of the homogeneous materials used in the component is below the limit on the acceptable amount stipulated in the GB/T 26572.

×: 表示该有害物质至少在该部件的某一均质材料中的含量超出 GB/T 26572 规定的限量要求。
Denotes that the amount of the hazardous substance contained in any of the homogeneous materials used in the component is above the limit on the acceptable amount stipulated in the GB/T 26572.